

ISHAN BHARDWAJ

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Quantitative finance graduate student with experience building end-to-end trading systems, predictive ML models, and automated analytical pipelines on large-scale financial datasets. Skilled in systematic strategy development, backtesting engine design, walk-forward validation, live broker integration (IBKR API), statistical modeling, and equity research. Targeting quant research, systematic trading, and data science in finance roles.

EDUCATION

University of Minnesota — Twin Cities	M.S. Financial Mathematics	Expected Dec 2026
KIIT University	B.Tech, Computer Science	Jul 2020 – May 2024

SKILLS

Trading & Quant	Systematic strategy development, backtesting engine design, walk-forward IS/OOS validation, IBKR API (ib_insync), ES/NQ/CL futures, options pricing, Monte Carlo methods
ML & Data	Python (pandas, numpy, scikit-learn, XGBoost, TensorFlow/Keras), SHAP, pyarrow, asyncio, time-series forecasting, feature engineering, SQL-style data processing
Finance	DCF valuation, SOTP, scenario analysis, equity research, credit risk modeling, logistic regression, KPI dashboards, Excel (advanced), n8n workflow automation, REST APIs

WORK EXPERIENCE

Equity Research Analyst Intern	Jun 2025 – Aug 2025
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Atlasio Capital Partners

- Built predictive financial models across 40+ companies, analyzing revenue, cost structures, and leverage ratios to identify margin inefficiencies of 300–500 bps across target sectors.
- Designed automated data workflows integrating financial datasets with validation pipelines, reducing manual data inconsistencies by 25% and improving downstream model reliability.
- Developed scenario-based pricing and cost sensitivity models across 10+ assumptions, quantifying tradeoffs between growth, margins, and credit risk; Python pipeline automation improved research turnaround by 20%.
- Delivered structured model-driven insights for senior stakeholders, translating complex analytical outputs into actionable investment recommendations across the coverage universe.

Quantitative Analyst Intern	Jan 2024 – Mar 2024
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Mudraksh & McShaw

- Analyzed datasets exceeding 500,000 records to identify statistical trends, cost drivers, and performance anomalies impacting business outcomes across multiple product lines.
- Built automated data pipelines and workflow triggers reducing manual processing time by 20–25%; developed KPI dashboards tracking 15+ operational and financial metrics for management reporting.
- Implemented data cleaning, feature engineering, and model validation workflows to improve modeling accuracy, scalability, and reporting efficiency across semi-automated analytics pipelines.

Trade Support Analyst Intern	Nov 2023 – Dec 2023
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AlgoBulls

- Processed and validated 1,000+ daily transactions across financial systems, ensuring near 100% accuracy and identifying discrepancies in transaction and pricing data to improve reporting reliability.
- Supported structured data workflows across 10,000+ transactions enabling audit readiness and regulatory compliance; contributed to automation initiatives reducing manual handling by 10–15%.

PROJECTS

CME Systematic Futures Trading System — CME Institute Competition, June 2026 | github.com/ISHAN217/CME-Futures-Trading-System

Python · pandas · numpy · pyarrow · asyncio · IBKR API (ib_insync) · Custom bar-level backtesting engine

- Built an end-to-end automated futures trading system covering IBKR live data acquisition, bar-level backtesting with fill verification, and live signal generation across 5 independent strategies; achieved OOS Sharpe of 4.85, IS/OOS ratio of 1.34x, and profitability in all 5 held-out years (2022–2026).
- Conducted fill validity audit revealing 70–80% of backtested entries were unfillable in practice and corrected win rate from 79.9% to an honest 53–60%; built 38-check test suite verifying P&L to the cent, IS/OOS ratio, and year-by-year profitability across all systems.
- Explored 15 signal types across ES, NQ, and CL futures; rejected 11 after OOS failure with documented reasons, retaining only signals with positive Sharpe across all held-out years — demonstrating rigorous research process over curve-fitting.

MISO Day-Ahead LMP Forecasting — Directed Research, University of Minnesota, May 2026 | github.com/ISHAN217/MISO-LMP-Forecasting

Python · scikit-learn · XGBoost · SHAP · pandas · matplotlib · seaborn

- Built a full ML forecasting pipeline testing whether net load and capacity stress features improve day-ahead electricity price prediction in MISO LRZ4; XGBoost and Ridge achieved RMSE \$11.56/MWh and 75–76% directional accuracy versus 36% for the naive prev-hour baseline — a 2x improvement.
- Engineered 23 features across 6 categories including a novel stress proxy (net load / rolling 30-day peak load) as an early-warning signal for high-price periods; SHAP analysis confirmed net load as the strongest predictor as renewable penetration grows across the MISO footprint.

MP Materials (NYSE: MP) — Strategic Equity Analysis | Airline Health Score (AHS+) Framework | github.com/ISHAN217/AHS

DCF · Scenario Analysis · Peer Comps · Rare Earth Markets | Python · DOT Data (DB1B, T-100, Form 41) · Statistical Modeling

- MP Materials: 12-month price target of \$42.50 from \$22.01 (+93%) achieved ahead of schedule via DoD procurement milestone; covered NdPr market dynamics, vertical integration strategy (Phases I–III), and EV/wind/defense demand with scenario-based DCF and peer benchmarks.
- AHS+: fuel-neutral composite scoring system for 260+ U.S. carriers across 100 months of DOT data (2017–2025) integrating cost efficiency, load factors, and revenue metrics; ranking consistency $\rho = 0.9997$, applied to equity research screening and operational benchmarking.